

上海交通大学试卷 (A 卷)

(2021 至 2022 学年 第 2 学期)

班级号 _____ 学号 _____ 姓名 _____

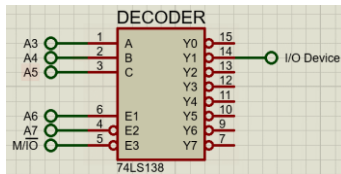
课程名称 计算机组成 成绩 _____

I. Single choice (20%)

1. About serial data communication, which of the following statement is correct? ()

- A. In synchronous serial communication, the sender and the receiver are synchronized by start bit.
- B. In asynchronous serial communication, each character uses only one stop bit.
- C. There is no idle state exists in asynchronous serial communication.
- D. Parity bit can be used in asynchronous serial communication.

2. Suppose the address lines A7~A0, which of the following address can activate I/O Device? ()



- A. 4CH
- B. 04H
- C. 40H
- D. 88H

3. About interrupts in 8086 systems, which of the following statement is correct? ()

- A. Hardware interrupts will be disabled by STI instruction.
- B. NMI interrupt is maskable.
- C. When the CPU is executing the ISR of interrupt N, the IF before the interrupt is saved in stack segment.
- D. 8255 is a chip for managing the interrupt request.

4. Suppose the stack is after the “ret” of the called proc is called. What is the CS:IP after return? ()

1ffb	1F
1ffc	00
1ffd	CB
1ffe	1F
1fff	N/A

- A. 001FH:1FCBH
- B. 1FCBH: 001FH
- C. 1F00H:CB1FH
- D. CB1FH: 1F00H

5. In the following operations, which would result in the value of carry flag being 0? ()

- A. 1000H-2000H B. FFFFH+FFFFH C. 6000H-3000H D. 2134H+E4C6H

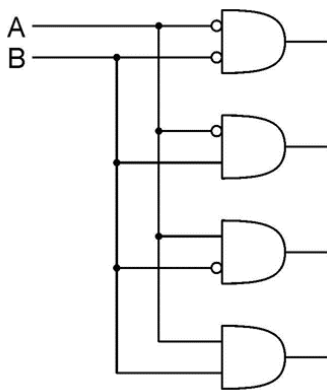
6. Suppose there are 64 SRAM chips and all of them are 256K×4bits. Which of the following SRAM memory can be formed and what is the number of address lines required for this memory (assume byte-addressable)? ()

- A. 1M×64bits SRAM, 26 B. 1M×8bits SRAM, 23 C. 1M×64bits SRAM, 23 D. 2M×8bits SRAM, 20

7. About interrupts in 8086 systems, which of the following statements is correct? ()

- A. Immediately after entering an ISR, the CPU can still respond to maskable interrupt requests.
B. The CPU can respond to an interrupt at any stage of the instruction pipeline.
C. Interrupt does not require the support from the hardware.
D. One interrupt can be further interrupted by other interrupts.

8. What is the output when AB=11? ()



- A. 0001 B. 1110 C. 0111 D. 1000

9. For 8253, the control word for configuring counter 1 in mode 3 with BCD mode and a single byte initial counter value is ().

- A. d7H B. 57H C. 55H D. 45H

10. Which segment register is used in the instruction MOV AL [BP]+3? ()

- A. CS B. DS C. SS D. ES

11. While calling DOS interrupt 21H, if you want to exit to DOS, the value in AH should be set to ().

- A. 09H B. 4CH C. 25H D. 26H

12. In an 8086 system, BX value is 0B52CH, after executing which following instruction BX value will be 0A965H ().

- A. ROL BX, 2 B. ROR BX, 2 C. ROL BX, 3 D. ROR BX, 3

13. Which of the following 8086 instruction could lead to the values of control signals $\overline{RD}$, $M/\overline{IO}$, $\overline{BHE}$ are 1,1,0 respectively? ()

- A. MOV AL, [2001] B. IN AH, 98H C. MOV [2000], AH D. None of the above

14. Which component does not belong to Von Neumann architecture? ()

- A. control unit B. input device C. memory D. disk

15. In an 8086 system, 0A3H, 0FFH, 12H and 2BH are stored starting from address 00080H. After executing 'INT 20H', the value of CS is ().

- A. 00A3H B. 12A3H C. 0FFA3H D. 2B12H

16. What is the address of the first instruction to be executed after the 8086 CPU is reset? ()

- A. 0FFFFH:0000H B. 0FFFFH:0FFFFH C. 0000H:0000H D. 0FFFFH:0FFFEH

17. Which following instruction would not change the flag register? ()

- A. AND B. ADD C. NOT D. CMP

18. If "DATA1 DW 07D3H" and "DATA2 DB 45H", after executing "MOV AX, DATA1" and "DIV DATA2", which registers would be used to store the quotient and the remainder, respectively? ()

- A. AH, AL B. AX, DX C. DX, AX D. AL, AH

19. The use of hierarchical memory system for computer is to ().

- A. easily manage the storage resource
B. solve the contradiction among capacity, price, and access speed
C. save a large amount of data
D. provide more choices to store data for programmers

20. Assume that 1 stop bit, 1 odd parity bit, and 7-bit character are used in an asynchronous transmission. If we want to send 125 characters per second, what is the baud rate? ()

- A. 1000 bps B. 1125 bps C. 1250 bps D. 1375 bps

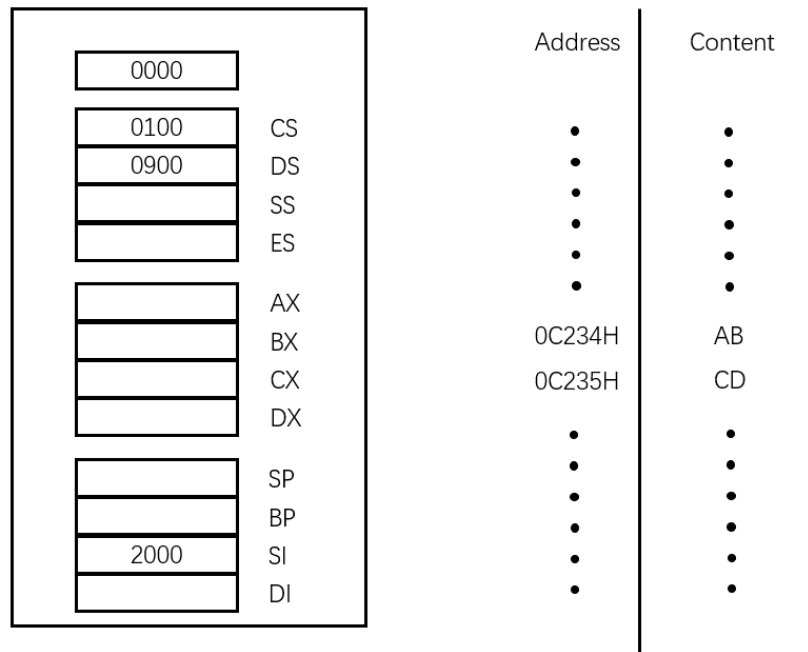
II. Fill in Blanks (20%)

1. Given that INT 21H is interrupt service for DOS, the corresponding interrupt vector consists of a CS and an IP value, which are physically located at _____H and _____H, respectively; when entering the ISR of INT 21H, IF = _____, which _____(allows/forbids) an INTR interrupt and _____(allows/forbids) an NMI interrupt.
2. 8086 adopts _____IO addressing scheme, in which _____and _____ instructions are used for memory and IO accesses, respectively, and CPU indicates either case with the _____ pin.
3. The addressable unit of the primary memory in a computer system is usually a _____; the word size of the system is determined by _____, for instance, the word size of an 8088 CPU is _____bits; and the address of an aligned word satisfies the condition that _____, for example, in an 8086 system, the address of an aligned word is _____(odd/even).
4. There are three main IO techniques, among which _____requires the minimal CPU intervention; _____ is the simplest; and _____is suitable for a computer system with multiple IO devices.
5. When building a memory module with memory chips, word extension is needed if _____ and after the extension each chip has _____ (the same/different) address range. In contrast, bit extension is required if _____.

III. Answer Questions (25%)

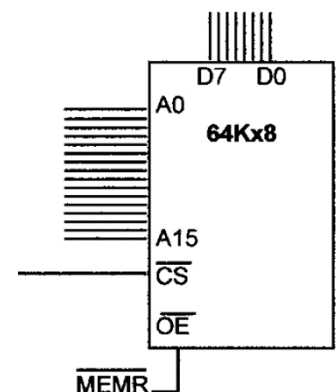
1. Suppose there are four input signals S_1, S_2, S_3, S_4 and one output signal O_1 . You need to design a digital circuit to let the output value $O_1 = (S_1 \text{ XOR } S_2) \text{ AND } (\text{NOT } S_3 \text{ OR } S_4)$ Please complete the truth table and draw the digital circuit. (6%)

2. Consider 8086 is executing the instruction MOV CX, [SI]+1234H and following system state (12%)



- a) Please describe the procedure of CPU computing the physical address of the operand and determine the final value in CX. (4%)

- b) Suppose the memory chip has 15 address lines and 8 data lines, what is the decoding circuitry of the memory chip that stores the desired data. Please draw the circuitry that connected to the chip selection pin. (4%)



- c) Please describe what happens if the offset 1234H becomes odd address. (4%)

3. Convert the following C program into 8086 assembly codes, assuming that int type is 16-bit. (7%)

```
void main() {  
    int array[10];  
    int n = 10;  
    f(array, n);  
}  
  
void f(int [] a, int n) {  
    for (int i = 0; i<n; i++) {  
        if (i<5)  
            a[i] = i;  
        else  
            a[i] = -i;  
    }  
}
```

IV. Experiments (35%)

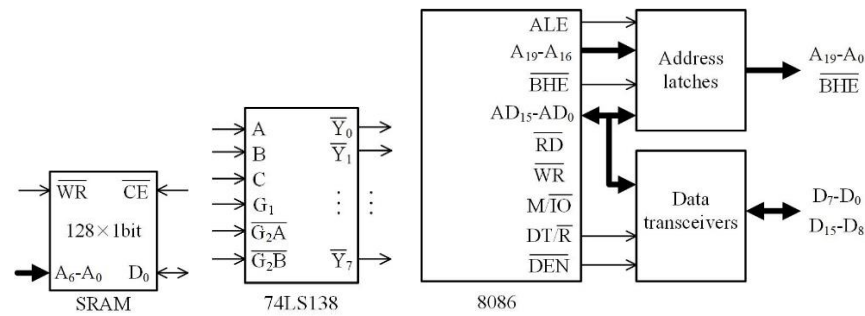
1. Two computers exchange data with each other via serial communication, where a frame is defined with 7-bit character, odd parity and 2 stop bits. Please answer the following questions. (8%)

(1) Draw the TxD signal for transmitting a character 55H. (4%)

TxD ———

(2) Explain why only one character is transmitted in asynchronous serial communication and why the frequency of the receiver clock (RxC) is N times that of the transmitter clock (TxC). (4%)

2. Design an 8086 memory module of 8KB, using 128×1-bit SRAM chips. The memory module is assigned to the lowest address space and both 8-bit and 16-bit accesses are allowed. (12%)

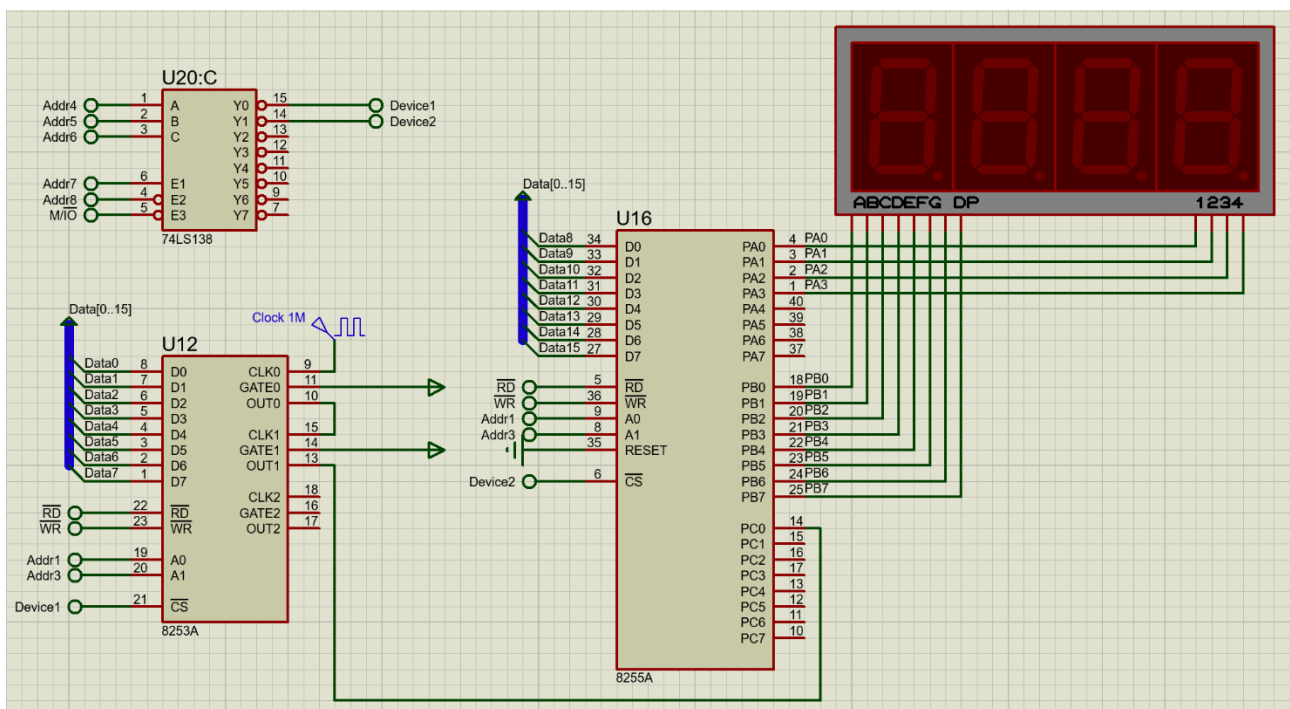


(1) Explain how 8-bit and 16-bit accesses are supported in 8086 systems; (3%)

(2) Explain how to design the desired address decoding scheme; (4%)

(3) Given the relevant signals of the SRAM chip, 74LS138 chip and the 8086 CPU, depict the memory organization scheme and how to connect the module to the system. (5%)

3. Given the following diagram, write a program to display a cumulative number on the LED digits. The initial number is “0000” and the number is incremented by 1 every 1 second and displayed. (15%)



(1) Define a procedure to initialize 8253; (3%)

(1) Define a procedure to initialize 8255; (3%)

(3) Define a procedure to check the falling edge on PC0; (3%)

(4) Complete the program to achieve the requirement.

```
DATA SEGMENT
; SEGTAB is the codebook for displaying 0-F
SEGTAB DB 3FH,06H,5BH,4FH,66H,6DH,7DH,07H,7FH,6FH,77H,7CH,39H,5EH,79H,7EH;
; You can define your variable if needed
```

```
DATA ENDS
CODE SEGMENT
    ASSUME CS:CODE,DS:DATA
START: MOV AX, DATA
        MOV DS, AX
; Write your code here and you can call the above defined procedures
```

CODE ENDS

END START